

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)	(i)		PbBr ₂		1		1		1
		(ii)		liquid neutral answer - molten	1			1		
		(iii)		bromine / Br ₂ accept Br ignore any reference to molten do not accept bromide / Br ⁻		1		1		1
		(iv)		$\text{Pb} + 2\text{e}^- \longrightarrow \text{Pb}^{2+}$ ☐ $\text{Pb}^{2+} - 2\text{e}^- \longrightarrow \text{Pb}$ ☐ $\text{Pb}^{2+} + 2\text{e}^- \longrightarrow \text{Pb}$ ☒ $\text{Pb} - 2\text{e}^- \longrightarrow \text{Pb}^{2+}$ ☐		1		1		
	(b)	(i)		coke (1) oxygen (1) limestone (1)	3			3		
		(ii)		B accept 2Fe + 3CO ₂		1		1	1	

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(c)	(i)		mild steel		1		1		
		(ii)		brittleness increases			1	1		
		(iii)		award (1) for any of following malleable easily shaped easy to bend do not accept ductile / soft / strong / hard			1	1		
		(iv)		A (1) award (1) for either of following - contains two different types of atoms / contains two elements / contains iron and carbon (atoms) - B only has one type of atoms and C has three types of atoms neutral answers contains two atoms / contains different atoms		1 1		2		
				Question 4 total	4	7	2	13	1	2